

#

Dynamics

→ Dynamics is the study of motion of body with ~~the~~ the cause of its motion.

• Inertia

→ The Tendency of a body to continue its state of rest or a uniform motion in a straight line unless an external force is applied to it is called inertia.

Types of Inertia

① Inertia of Rest GSS_Notes

- The tendency of a body to continue its state of rest unless an external force is applied to it is called Inertia of Rest.

② Inertia of motion

- The tendency of a body to continue its state of uniform motion in a straight line unless an external force is applied to it is called inertia of motion.

Q. When a bus suddenly starts to move then the passengers tend to fall backward. Why?

Ans. When a bus suddenly starts to move then lower portion of the passengers comes into motion along with the bus, but upper

portion of the passengers tend to remain at rest due to inertia of rest. As a result, the passengers tend to fall backward.

Q. When a bus is suddenly stopped, then the passengers tend to fall forward. Why?

Ans. When a bus suddenly stops, ~~the lower portion~~ then lower portion of the passengers come in rest along with the bus but upper ~~part~~ portion of the passengers tend to remain in motion due to inertia of motion. As a result, the passengers tend to fall forward.

* Note:

- Inertia is directly proportional to the mass of a body
- Newton's First Law of motion

↳ It states that, "Everybody in this universe continues its state of rest or of uniform motion in a straight line unless an external force is applied to it."

According to this law, Force is an external agent which changes or tries to change the state of a body from rest to mo-

tion or motion to rest.

It is qualitative definition of force.

• Linear Momentum ($\vec{p}$)

↳ The total quantity of motion contained in a body is called linear momentum.

The linear momentum of a body of mass (m) moving with velocity ($\vec{v}$) is given by:

$$\begin{aligned}\vec{p} &= m\vec{v} \\ &= \text{kg ms}^{-1} \\ &= [\text{MLT}^{-1}]\end{aligned}$$

• Newton's Second Law of motion

GSS_Notes

↳ It states that, "The rate of change of linear momentum of a body with time is directly proportional to the magnitude of applied force and the change in linear momentum takes place in the direction of applied force.

$$\text{i.e.} \quad \frac{d\vec{p}}{dt} \propto \vec{F}$$

$$\text{Conversely, } \vec{F} \propto \frac{d\vec{p}}{dt}$$

$$\therefore \vec{F} = k \cdot \frac{d\vec{p}}{dt}$$

Where k is proportionality constant

and $k=1$

$$\therefore \vec{F} = \frac{d\vec{P}}{dt} \longrightarrow (1)$$

Hence, According to Newton's second Law of motion, Force is defined as the rate of change of linear momentum of a body with time.

If m be the mass and $\vec{v}$ be the velocity of a body then linear momentum of the body is given by

$$\vec{P} = m\vec{v}$$

Then from eqn (i),

$$\vec{F} = \frac{d(m\vec{v})}{dt}$$

$$\text{or, } \vec{F} = m \frac{d\vec{v}}{dt}$$

$$\therefore \vec{F} = m\vec{a} \longrightarrow (2)$$

Where $\vec{a} = \frac{d\vec{v}}{dt}$ is the acceleration of the body.

Q 1) A cricket ball of mass 145 gm is moving with velocity of 14 m/s and is being hit by a bat so that the ball is turned back with a velocity 22 m/s. The force of blow acts on the ball for 0.015 sec. Find the average Force exerted by the bat on the ball.

→ Soln,

$$m = 145 \text{ gm} = \frac{145}{1000} \text{ kg} = 0.145 \text{ kg}$$

$$u = 14 \text{ m/s}$$

$$v = -22 \text{ m/s}$$

$$t = 0.015 \text{ sec}$$

$$F = ?$$

Now,

$$F = ma$$

$$= m \times \left(\frac{v - u}{t} \right)$$

$$= 0.145 \times \left[\frac{-22 - 14}{0.015} \right]$$

$$= 0.145 \times \left[\frac{-36}{0.015} \right]$$

$$= -348 \text{ N}$$

$$= 348 \text{ N (in Magnitude)}$$

Q 2) A vehicle of mass 500 kg is moving with a speed of 10 m/s. Sand is dropped into it at the rate of 10 kg/min. What Force is needed to keep the vehicle moving with uniform speed?

→ Soln,

$$m_v = 500 \text{ kg}$$

$$V = 10 \text{ m/s}$$

$$\frac{dm_s}{dt} = 10 \text{ kg/min} = \frac{10}{60} \text{ kg/sec} = \frac{1}{6} \text{ kg/sec}$$

$$F = ?$$

Now,

$$F = \frac{dP}{dt}$$

$$= \frac{d}{dt} [(m_v + m_s) V]$$

$$= V \frac{d}{dt} (m_v + m_s) \quad [\because V \rightarrow \text{constant}]$$

$$= V \left[\frac{dm_v}{dt} + \frac{dm_s}{dt} \right]$$

$$= V \left[0 + \frac{dm_s}{dt} \right]$$

$$= V \times \frac{dm_s}{dt} \quad \text{GSS_Notes}$$

$$= 10 \times \frac{1}{6}$$

$$= 1.67 \text{ N}$$

- ③ The mass of gas emitted from the rear of toy rocket is initially 0.2 kg/s. If the speed of the gas relative to the rocket is 40 m/s and mass of rocket is 4 kg, what is the initial acceleration of the rocket.

→ Soln,

$$\left(\frac{dm}{dt} \right)_g = 0.2 \text{ kg/s}$$

$$V_g = 40 \text{ m/s}$$

$$m_r = 4 \text{ kg}$$

$$a_r = ?$$

We know that,

$$\begin{aligned} F_g &= \left(\frac{dp}{dt} \right)_g \\ &= \left(\frac{d(mv)}{dt} \right)_g \\ &= v_g \left(\frac{dm}{dt} \right)_g \\ &= 40 \times 0.2 \\ &= 8 \text{ N} \end{aligned}$$

Now,

According to third Law

$$F_r = F_g$$

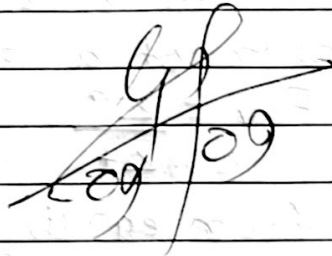
$$\therefore F_r = 8 \text{ N}$$

Then,

$$F_r = m_r a_r$$

$$\text{or } 8 = 4a_r$$

$$\therefore a_r = 2 \text{ m/s}^2$$



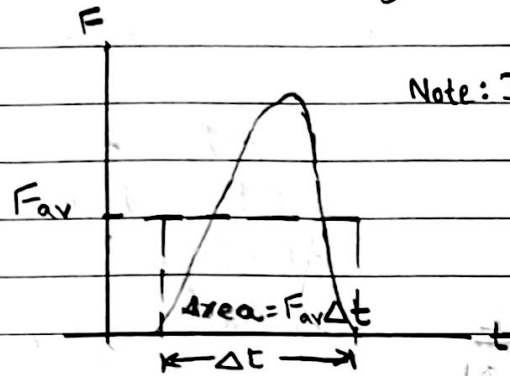
• Impulse

↳ Impulse is the measure of effect of large force acting on a body for a very short time. Mathematically, impulse is given by the product of average force acting on a body and the time for which the force is applied.

i.e. Impulse = average force $\times$ time

$$\therefore I = F_{av} \times \Delta t \rightarrow (i)$$

In impulse, applied force rises sharply from zero to maximum and falls to zero in a very short interval of time as shown in figure.



Note: Impulse = Area below the curve

In the figure, the area of shaded region gives impulse.

According to the Newton's second law, force is the rate of change in linear momentum of a body.

$$\text{i.e. } F_{av} = \frac{\Delta P}{\Delta t}$$

Then from eqn (i), we get

$$I = \frac{\Delta P}{\Delta t} \times \Delta t$$

$$\therefore I = \Delta P$$

i.e. Impulse = change in linear momentum.

Give reason

i) A cricket player moves his hands backward while catching a cricket ball.

ii) Ceramics are wrapped with straw or paper before transporting them.

iii) A person feels more hurt when falls on cemented floor than on sand.

iv) Vehicles are provided with spring system.

Solution

GSS_Notes

i) When a cricket player moves his hand ~~while~~ backward while catching the cricket ball then the time of impact between the ball and the hands increases. According to ~~Newton's~~ Newton's second Law of motion, Applied Force is inversely proportional to the time of impact so less force is applied and hence, the player feels less hurt while catching ball.

ii) Ceramics are wrapped with straw and paper because it increases the time during which a force acts when they are shaken or hit. This reduces the force on the ceramic and helps prevent it from breaking.

iii) A person feels more hurt when falling on a cemented floor because the stopping time is very short, so a large force acts on the body. Sand increases the stopping time by cushioning the fall, which reduces the force and causes less injury.

iv) Vehicles are provided with a spring system because springs increase the time of impact when the vehicle passes over bumps or uneven roads. This reduces the force transmitted to the passengers, making the ride smoother and preventing damage.

• Newton's third law of motion

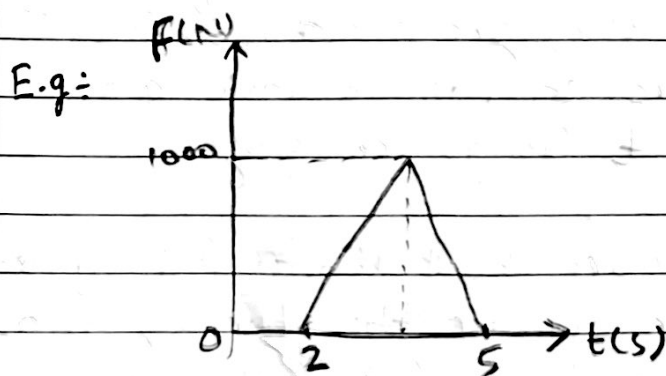
→ It states that, "To every action there is an equal and opposite reaction."

Give reason

i) When bullet is fired from a gun, the gun recoils backward. Why?

ii) We push water backward while swimming. Why?

iii) Action and reaction are equal and opposite but they do not cancel each other. Why?



Here,

$$\begin{aligned}\text{Impulse} &= \text{Area of } \triangle \\ &= \frac{1}{2} b \times h\end{aligned}$$

$$= \frac{1}{2} \times 3 \times 1000$$

$$= 1500 \text{ Ns}$$

Solution

- i) When bullet is fired from a gun, then the bullet moves in forward direction with certain force which is action. According to ~~the~~ Newton's third law of motion, "to every action There is equal and opposite reaction." so as a reaction the gun recoils backward.
- ii) We push water backward ~~in~~ with certain force which is action. According to Newton's third law of motion, "To every action there is equal and opposite reaction." so as a reaction we go forward.
- iii) Action and reaction are equal ~~for~~ ~~two~~ ~~different~~ ~~object~~ and opposite but they do not cancel each other because they act on two different bodies. Forces cancel only when they act on the same body in opposite directions.

• Principle of Conservation of Linear Momentum

↳ It states that, "If no external force is applied on a system, the total linear momentum of the system remain constant."

i.e. in a colliding system,

Total linear momentum before collision = total linear momentum after collision.

* Proof from Newton's second law of motion

According to the Newton's second law of motion,

Force = Rate of change of linear momentum with time

$$\text{i.e. } F = \frac{dP}{dt}$$

GSS_Notes

if no external force is applied on a system i.e. $F = 0$ then

$$\frac{dP}{dt} = 0$$

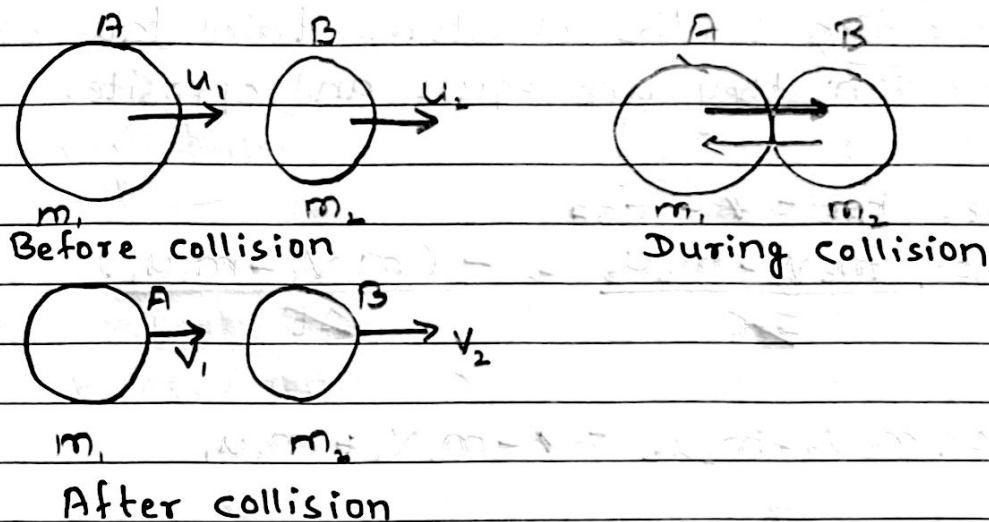
Integrating both sides we get
 $P = \text{Constant}$

This proves principle of conservation of linear momentum.

* Proof from Newton's third law of motion

Consider two bodies A and B of masses m_1 and m_2 are moving along the same straight line and in the same direction with velocities u_1 and u_2 respectively such

that $u_1 > u_2$. Suppose the bodies collide for a short time Δt and after collision they move with different velocities v_1 and v_2 as shown in figure.



During collision, the body A exerts force on the body B and the body B also exerts force on the body A.

Now, the force exerted by A on B is given by

$$F_{AB} = \frac{\text{Change in linear momentum of B}}{\text{time of collision}}$$

$$= \frac{m_2 v_2 - m_2 u_2}{\Delta t} \quad \text{--- (i)}$$

Similarly, force exerted by B on A is given by

$$F_{BA} = \frac{\text{Change in linear momentum of A}}{\text{time of collision}}$$

$$= \frac{m_1 v_1 - m_1 u_1}{\Delta t} \rightarrow (ii)$$

Here, F_{AB} and F_{BA} are action and reaction. According to the Newton's third law of motion, they are equal and opposite.

i.e. $F_{AB} = -F_{BA}$

$$\text{or, } \frac{m_2 v_2 - m_2 u_2}{\Delta t} = - \frac{(m_1 v_1 - m_1 u_1)}{\Delta t}$$

$$\text{or, } m_2 v_2 - m_2 u_2 = -m_1 v_1 + m_1 u_1$$

$$\text{or, } m_2 v_2 + m_1 v_1 = m_1 u_1 + m_2 u_2$$

$$\therefore m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

i.e. Total linear momentum before collision = Total linear momentum after collision.

This proves principle of conservation of linear momentum.

* Newton's Second Law of motion as a real law

→ Both the Newton's First law of motion and third law of motion can be derived from the second law of motion, so New-

ton's second law of motion is also known as real law of motion.

① Derivation of Newton's First law of motion from second law

According to Newton's second Law of motion, we have

$$F = ma$$

If force is not applied on a body

i.e. $F = 0$ then,

$$ma = 0$$

$$\text{or, } a = 0$$

$$\text{or, } \frac{v - u}{t} = 0$$

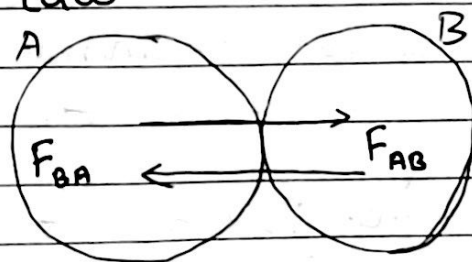
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$$\text{or, } v - u = 0$$

$$\therefore v = u$$

This shows that if no external force is applied on a body then the body continues its state of rest or of uniform motion in a straight line. This is Newton's First law of motion.

② Derivation of Newton's Third Law of motion from second law



Consider two bodies A and B undergo collision for a small interval of time Δt . During collision the body exerts force F_{AB} on the body B which is action and the body B also exerts force F_{BA} on the body A which is reaction.

According to Newton's second law of motion, we have

$$F_{AB} = \frac{\text{change in linear Momentum of B}}{\text{time of collision}}$$

$$\text{or, } F_{AB} = \frac{\Delta P_B}{\Delta t}$$

$$\therefore \Delta P_B = F_{AB} \Delta t \rightarrow \textcircled{1}$$

Similarly,

$$F_{BA} = \frac{\Delta P_A}{\Delta t}$$

$$\therefore \Delta P_A = F_{BA} \Delta t \rightarrow \textcircled{2}$$

Then, total change in the linear momentum of the system is given by

$$\Delta P = \Delta P_A + \Delta P_B$$

$$\therefore \Delta P = F_{BA} \Delta t + F_{AB} \Delta t \rightarrow \textcircled{3}$$

According to principle of conservation of linear momentum, the total momentum of the system remains constant.

$$\text{i.e. } P = \text{constant}$$

$$\Delta P = 0$$

$$\text{or, } F_{BA} \Delta t + F_{AB} \Delta t = 0$$

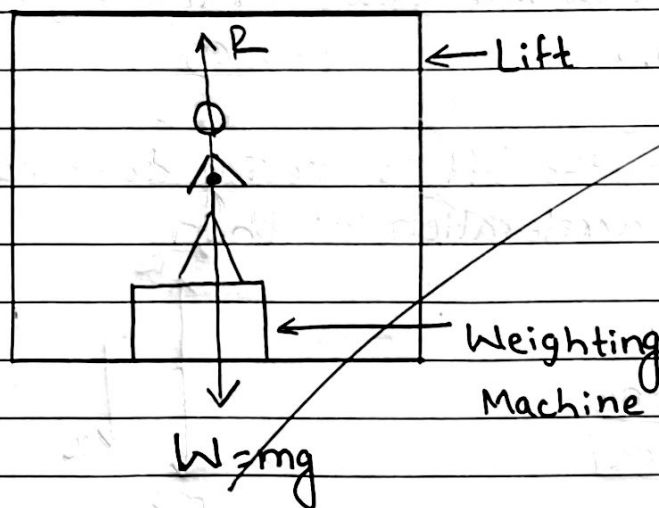
$$\text{or, } F_{AB} \cancel{\Delta t} = -F_{BA} \cancel{\Delta t}$$

$$\therefore F_{AB} = -F_{BA} \rightarrow (4)$$

This shows that action and reaction are equal and opposite. This is Newton's third law of motion.

• Free body Diagram

① A man in a lift



Consider a man of mass m is standing on a weighting machine inside lift as shown in figure.

The different ~~form~~ forces acting on the man are:

- ① True weight (w) = mg , acting downward.
- ② Normal reaction (R), acting upward

Here, R is the weight of man shown by the weighting machine called apparent weight of man.

Case (I) - When the lift is at rest or moving with constant velocity then,

$$F_{\text{net}} = ma$$

$$\text{or, } F_{\text{net}} = 0 \quad [\because \text{At rest or constant velocity, } a=0]$$

$$\text{or, } R - mg = 0$$

$$\therefore R = mg$$

i.e. weighting machine shows true weight of the man.

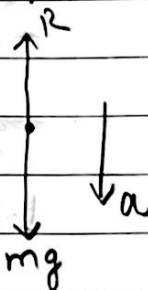
GSS_Notes

Case (II) - When the lift is moving downward with acceleration 'a' then

$$F_{\text{net}} = ma$$

$$\text{or, } mg - R = ma$$

$$\therefore R = mg - ma$$



Here, $R < mg$ i.e. apparent weight of man appears less than the true weight.

If the lift is in free fall i.e. $a=g$ then

$$R = mg - mg$$

$$\therefore R = 0$$

i.e. At free fall, the apparent weight of man appears zero.

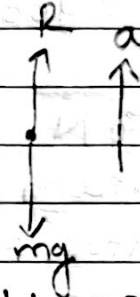
This condition is called weightlessness.

Case (III) - When the lift moves upward with acceleration 'a' then

$$F_{\text{net}} = ma$$

$$\text{or, } R - mg = ma$$

$$\therefore R = mg + ma$$



Here, $R > mg$ i.e. apparent weight appears greater than true weight of man.

Q. 3) A lift moves (i) up and (ii) down with an acceleration of 2 ms^{-1} . ~~Sand is dropped into it at the rate~~ In each case, calculate the reaction of the floor on a man of mass 50 kg standing in the lift.

→ Soln,

$$a = 2 \text{ ms}^{-1}$$

$$m = 50 \text{ kg}$$

$$R = ?$$

(i) When lift moves up.

$$R = mg + ma$$

$$= 50 \times 10 + 50 \times 2$$

$$= 500 + 100$$

$$= 600 \text{ N}$$

(ii) When lift moves down

$$\begin{aligned} R &= mg - ma \\ &= 50 \times 10 - 50 \times 2 \\ &= 500 - 100 \\ &= 400 \text{ N} \end{aligned}$$

Q.4 A 550 N physics student stands on a bathroom scale in an elevator. As the elevator starts moving the scale reads 450 N. Draw free body diagram of the problem and find the magnitude and direction of the acceleration of the elevator.

→ Soln,

$$W = mg = 550 \text{ N}$$

$$\therefore m = \frac{550}{10} = 55 \text{ kg}$$

$$R = 450 \text{ N}$$

$$a = ?$$

Now,

$$F_{\text{net}} = ma$$

$$W - R = ma$$

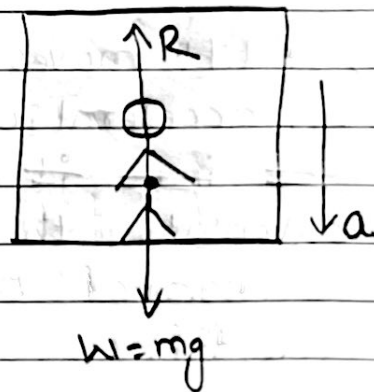
$$\text{or, } 550 - 450 = 55 \times a$$

$$\text{or, } 100 = 55a$$

$$\text{or, } a = \frac{100}{55}$$

$$\therefore a = 1.82 \text{ m/s}^2$$

Here $W > R$. So, the acceleration is downward.



② Mass - Pulley System.

Let, $m_1 < m_2$

For mass m_1

$$F_{\text{net}} = m_1 a$$

$$\therefore T - m_1 g = m_1 a \quad \text{--- (i)}$$

For mass m_2

$$F_{\text{net}} = m_2 a$$

$$\therefore m_2 g - T = m_2 a \quad \text{--- (ii)}$$

Adding (i) and (ii)

GSS_Notes

$$\begin{aligned} T - m_1 g &= m_1 a \\ + m_2 g - T &= m_2 a \\ \hline \end{aligned}$$

$$\begin{aligned} m_2 g - m_1 g &= m_1 a + m_2 a \\ \text{or, } (m_2 - m_1) g &= (m_1 + m_2) a \end{aligned}$$

$$\therefore a = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g$$

Again,

Dividing (i) by (ii)

$$\frac{T - m_1 g}{m_2 g - T} = \frac{m_1 a}{m_2 a}$$

$$\text{or, } m_1 \cdot m_2 g - m_1 T = m_2 T - m_1 \cdot m_2 g$$

$$\text{or, } m_1 \cdot m_2 g + m_1 \cdot m_2 g = m_2 T + m_1 T$$

$$\text{or, } (m_1 \cdot m_2 + m_1 \cdot m_2) g = (m_1 + m_2) T$$

$$2 m_1 \cdot m_2 g = T (m_2 + m_1)$$

$$\therefore T = \left(\frac{2 m_1 \cdot m_2}{m_1 + m_2} \right) g$$

Q.5. Two masses 7 kg and 12 kg are connected at the two ends of a light inextensible string that passes over a frictionless pulley. Using free body diagram method, find the acceleration of masses and the Tension in the string, when the masses are released.

→ Soln,

$$m_1 = 7 \text{ kg}$$

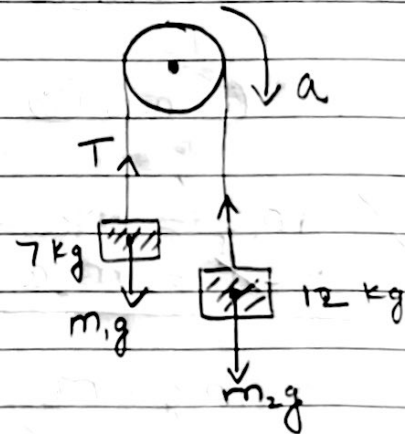
$$F_{\text{net}} = m_1 a$$

$$\text{or, } T - m_1 g = m_1 a$$

$$\text{or, } T - 7 \times 10 = 7 \times a$$

$$\text{or, } T - 70 = 7a$$

$$\text{or, } T = 70 + 7a \rightarrow \textcircled{i}$$



$$m_2 = 12 \text{ kg}$$

$$F_{\text{net}} = m_2 a$$

$$m_2 g - T = m_2 a$$

$$12 \times 10 - T = 12a$$

$$120 - T = 12a$$

$$T = 120 - 12a \rightarrow \textcircled{ii}$$

From $\textcircled{i}$ and $\textcircled{ii}$

$$70 + 7a = 120 - 12a$$

$$12a + 7a = 120 - 70$$

$$19a = 50$$

$$a = \frac{50}{19}$$

$$a = 2.63 \text{ m/s}^2$$

Now,

$$T = 70 + 7a$$

$$T = 70 + 7 \times 2.63$$

$$= 70 + 18.42$$

$$= 88.42 \text{ N}$$

$$\therefore T = 88.42 \text{ N}$$

$$a = 2.63 \text{ m/s}^2$$

~~Find tension in the string and acceleration of the masses from the given figure.~~

• Friction

→ The opposing force developed between two surfaces in contact when a body moves or tries to move over another body is called friction.

Types of friction

① Static Friction

- The force of friction developed between two surfaces in contact until the body is at rest is called static friction.

Static friction is always equal to the app-

lied force

> Limiting Friction

→ The Frictional force developed between two surfaces in contact when a body just starts to move over another body is called limiting friction.

OR.

→ The Maximum value of static friction is called Limiting friction.

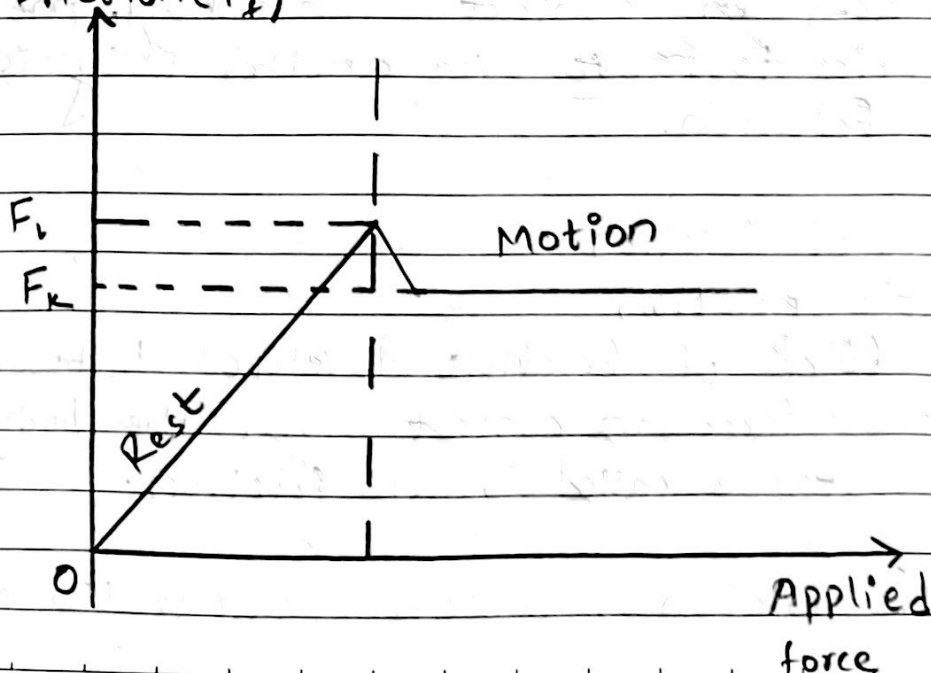
② Kinetic Friction / Dynamic Friction

→ The Frictional force developed between two surfaces in contact when a body is in motion over another body is called Kinetic Friction.

GSS_Notes

Kinetic Friction is always less than the limiting friction.

Friction (F_f)



• Laws of limiting friction

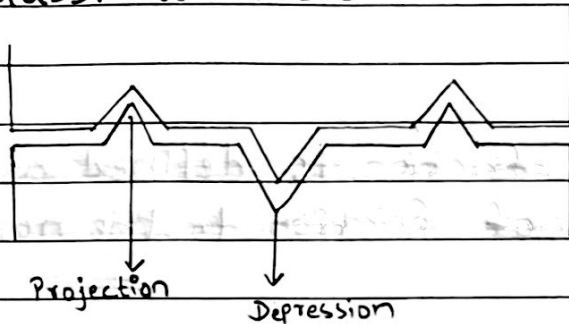
(i) Force of friction is directly proportional to the normal reaction.

$$\text{i.e. } f_r \propto R$$

(ii) Force of friction is independent to the surface area in contact.

• Cause of friction

(i) Classical view



→ When a body moves over another body then there is interlocking between the projection and depression of two surfaces which causes friction.

(ii) Modern view

→ When two smooth surfaces are in contact then there is attractive force called adhesive force between the molecules of two surfaces which causes friction.

→ It is also known as cold welding.

• ~~Law~~ Coefficient of friction

→ According to Law of solid friction, the force of friction is directly proportional to the normal reaction.

i.e. $f_f \propto R$

$$\therefore f_f = \mu R \rightarrow \textcircled{i}$$

Where μ is the proportionality constant which is called coefficient of friction.

From $\textcircled{i}$

$$\mu = \frac{f_f}{R}$$

Hence coefficient of friction is defined as the ratio of force of friction to the normal reaction.

• Angle of Friction

Here,

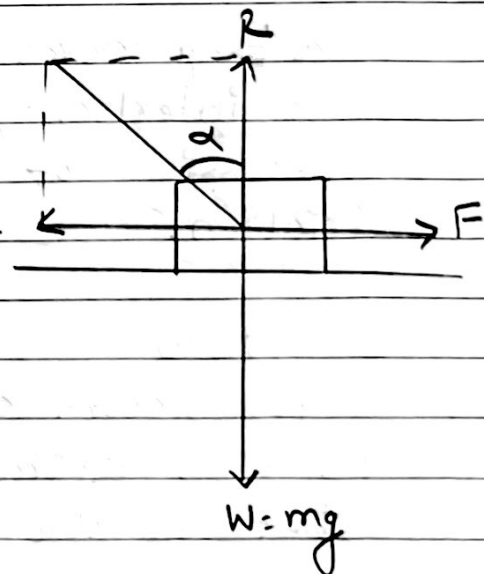
$W = mg \rightarrow$ Weight of body

$R =$ Normal reaction

$F =$ Applied Force

$f_f =$ Frictional Force

$\alpha =$ Angle of friction



The angle made by the resultant of frictional force and normal reaction with the normal reaction is called angle of

friction.

From figure,

$$\tan d = \frac{F_f}{R}$$

$$\therefore \tan d = \mu \rightarrow (i)$$

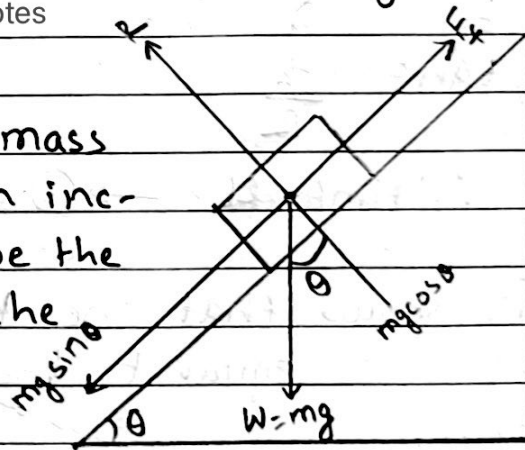
This shows that the tangent of angle of friction is equal to the coefficient of friction.

✓ Angle of Repose

→ The angle of inclination of an inclined plane at which a body placed on it just starts to slide down is called angle of repose.

GSS_Notes

Consider a body of mass m is placed on an inclined plane. Let θ be the angle of repose of the plane.



The different forces acting on the body are:

- (i) Weight (W) = mg
- (ii) Normal Reaction (R)
- (iii) Frictional Force (F_f)

The weight mg can be resolved into two rectangular components $mg \sin \theta$ and $mg \cos \theta$ as shown in fig.

Here, the component $mg \sin \theta$ is balanced by frictional force

$$\therefore mg \sin \theta = F_f \rightarrow (1)$$

And the component $mg \cos \theta$ is balanced by normal reaction.

$$\therefore mg \cos \theta = R \rightarrow (2)$$

Dividing (1) by (2)

$$\frac{mg \sin \theta}{mg \cos \theta} = \frac{F_f}{R}$$

$$\text{or, } \tan \theta = \frac{F_f}{R}$$

$$\therefore \tan \theta = \mu \rightarrow (3)$$

This shows that the tangent of angle of repose is equal to the coefficient of friction.

Note: Since, $\tan \alpha = \mu$ and $\tan \theta = \mu$

so,

$$\tan \alpha = \tan \theta$$

$$\therefore \alpha = \theta$$

This shows that the angle of friction (α)

and the angle of repose (θ) are numerically equal.

Q.6) An iron block of mass 10 kg rests on a wooden plane inclined at 30° to the horizontal. It is found that the Least force parallel to the plane which causes the block to slide up is 100N. Calculate the coefficient of friction between the two surfaces.

→ Soln,

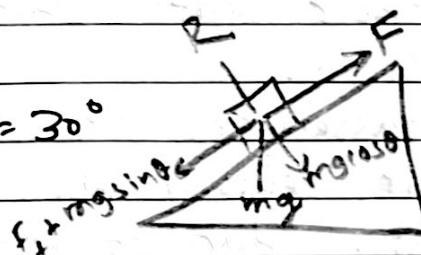
$$\text{mass}(m) = 10 \text{ kg}$$

$$\text{angle of inclination } (\theta) = 30^\circ$$

$$\text{Least force } (F) = 100 \text{ N}$$

Now,

~~downward~~ components
 ~~$mg \sin \theta$~~



$$F_f + mg \sin \theta = F$$

$$\text{or, } \mu R + mg \sin \theta = F$$

$$\text{or, } \mu mg \cos \theta + mg \sin \theta = F$$

$$\text{or, } g [\mu \times 10 \times \cos 30 + 10 \times \sin 30] = 100$$

$$\text{or, } g \left[\mu \times \frac{10\sqrt{3}}{2} + 10 \times \frac{1}{2} \right] = 100$$

$$\text{or, } g \left[\mu \times \frac{10\sqrt{3}}{2} \right] = 50$$

$$\text{or, } g \mu = \frac{50}{5\sqrt{3}} \quad \mu \times 5\sqrt{3} = 5$$

$$\text{or, } \mu = \frac{1}{\sqrt{3}}$$

$$\therefore \mu = \frac{1}{\sqrt{3}}$$

$$\therefore \mu = \frac{1}{\sqrt{3}}$$

$$\therefore \mu = \frac{1}{\sqrt{3}}$$

Conceptual short Questions

1. The Leaves fall when a tree is shaken.

→ According to Newton's First Law of motion, the leaves fall when a tree is shaken because tree comes in motion but leaves ~~try~~ tries to continue inertia of rest and falls down.

GSS_Notes

2. Why are passengers thrown forward from their seats when a speeding bus suddenly stops?

→ Passengers are thrown forward from their seats when a speeding bus suddenly stops due to inertia of motion. Passengers tries to continue inertia of motion but bus comes in rest and eventually passengers falls forward.

3. Why does a cricketer lower his hand while catching cricket ball?

→ Cricketer lower his hand while catching a cricket ball to increase time of impact. According to Newton's Second law, force is inversely proportional to time of impact. So force decreases while increasing time of impact and it strikes hand with less force.

4) Why it is difficult to run fast on sand?

→ Running on sand is hard as sand is not rigid. When the foot pushes sand backward, sand deforms and slips, so the frictional force and reaction force are small. As a result, less force is available for forward motion and more energy is lost, making running difficult.

5) Why does a rifle not kick as strongly as a light rifle using the

→ When a bullet is fired forward, the rifle must move backward so the total momentum remains zero.

$$m_b v_b = m_r v_r$$

If m_r is large, recoil velocity v_r becomes small. If m_r is small, recoil velocity v_r becomes large.

Therefore, a light rifle moves backward faster, giving a strong kick, while a heavy rifle recoils more slowly, giving a weaker kick.

7) A woman in an elevator lets go of her briefcase but it doesn't fall to the floor. How is the elevator moving?

→ When the woman releases her briefcase,

the only force on the briefcase is gravity (mg), so its acceleration downward is 'g' which means it is in freefall. So the elevator must be moving downward at the value of g.

GSS_Notes

⑦ In a physics Lab experiment, a 6 kg box is pushed across a flat table by a horizontal force F .

① If the box is moving at a constant speed of 0.35 m/s and the coefficient of kinetic friction is 0.12 , what is the magnitude of F ?

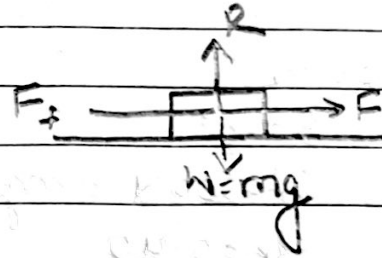
→ Soln,

$$m = 6 \text{ kg}$$

$$v = 0.35 \text{ m/s}$$

$$\mu = 0.12$$

$$F = ?$$



We know that,

$$F_f = F \quad [\because a = 0]$$

$$\text{or, } \mu R = F$$

$$\text{or, } \mu mg = F$$

$$\text{or, } 0.12 \times 6 \times 10 = F$$

$$\therefore F = 7.2 \text{ N}$$

ii) If the box is speeding up with a constant acceleration of 0.18 m/s^2 , what will be the magnitude of F ?

→ Soln,

$$a = 0.18 \text{ m/s}^2$$

$$F = ?$$

Now,

$$F - F_f = ma$$

$$\text{or, } F - 7.2 = 6 \times 0.18$$

$$\text{or, } F = 1.08 + 7.2$$

$$\therefore F = 8.28 \text{ N}$$

- ② Suppose you try to move a crate by tying a rope around it and pulling on the rope at angle of 30° above the horizontal. What is the tension required to keep the crate moving with constant velocity? [Weight (W) = 500 N , Coe. of dynamic friction $\mu_k = 0.40$]

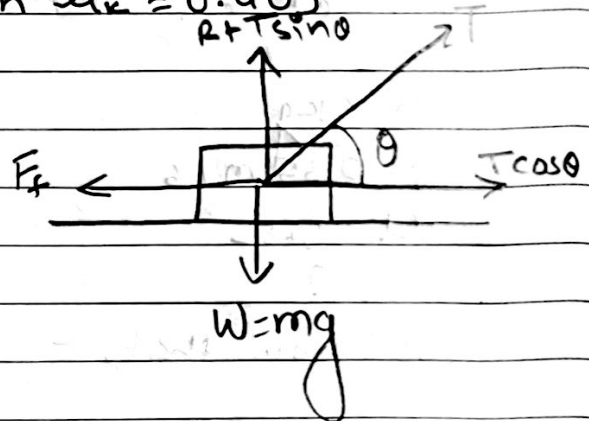
→ Soln,

$$\theta = 30^\circ$$

$$W = 500\text{ N} = mg$$

$$\mu_k = 0.40$$

$$T = ?$$



Now,

$$T \cos \theta = F_f$$

$$\text{or, } T \cos 30 = \mu R$$

$$\text{or, } T \frac{\sqrt{3}}{2} = 0.40 (mg - T \sin \theta)$$

$$\text{or, } T \frac{\sqrt{3}}{2} = 0.40 (500 - T \sin 30)$$

$$\text{or, } T \frac{\sqrt{3}}{2} = 0.40 \left(500 - \frac{T}{2} \right)$$

$$\text{or, } T \frac{\sqrt{3}}{2} = 200 - \frac{1}{2} \times 0.2 T$$

$$\text{or, } \left(\frac{\sqrt{3}}{2} + 0.2 \right) T = 200$$

$$\text{or, } T = 187.612$$

$$\therefore T = 187.61\text{ N} \underline{\underline{11.}}$$

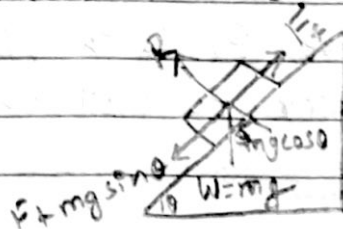
Q2) What would be the acceleration of a block sliding down an inclined plane that makes an angle of 45° with the horizontal if the coefficient of sliding friction between two surfaces is 0.3?

→ Soln,

$$\theta = 45^\circ$$

$$\mu = 0.3$$

$$a = ?$$



We know that,

$$mg \sin \theta - F_f = ma$$

$$\text{or, } mg \sin \theta - \mu R = ma$$

$$\text{or, } mg \sin \theta - \mu mg \cos \theta = ma$$

$$\text{or, } g \sin \theta - \mu g \cos \theta = a$$

$$\text{or, } 10 \times \sin 45^\circ - 0.3 \times 10 \times \cos 45^\circ = a$$

$$\text{or, } 10 \times \frac{1}{\sqrt{2}} - 0.3 \times 10 \times \frac{1}{\sqrt{2}} = a$$

$$\text{or, } a = 4.94 \text{ m/sec}^2$$

$$\therefore a = 4.94 \text{ m/sec}^2$$

Why it is easier to pull than to push?

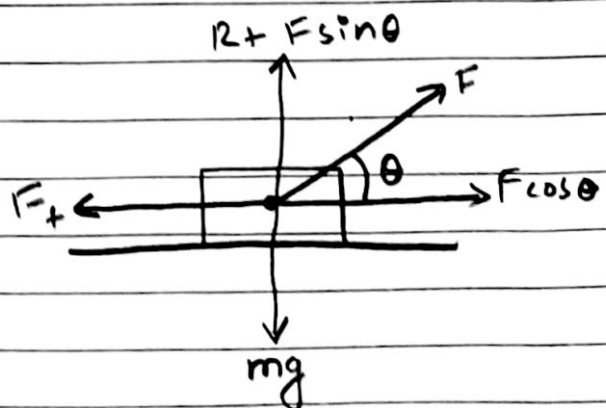
> Case (i) Pull

- Here,

$$F_f = \mu R$$

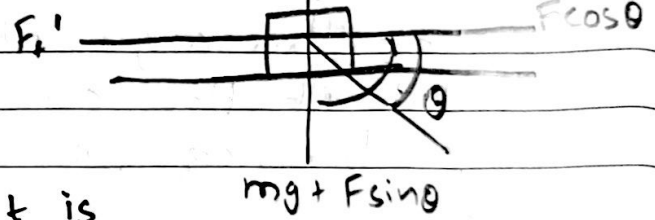
$$F_f = \mu (mg - F \sin \theta)$$

← Li



Case (ii) - Push

$$F_f' = \mu R$$
$$= \mu (mg + P \sin \theta)$$



Here, $F_f < F_f'$, so it is easier to pull than push

GSS_Notes